

**ANSH THAPA****B.TECH CSE****CU24250224****Experiment No. 4**

Frequency Domain Image Filtering using Fourier Transform

**Aim**

To implement image filtering in the frequency domain using the Fourier Transform and analyze the effectiveness of frequency-based filtering techniques for noise removal, image enhancement, and feature preservation using Python and OpenCV.

**Purpose**

Many image processing operations can be performed more efficiently in the frequency domain than in the spatial domain. The Fourier Transform converts an image from the spatial domain into its frequency components, enabling selective manipulation of low-frequency and high-frequency information. Frequency domain filtering is widely used in applications such as medical image enhancement, satellite image analysis, image restoration, and biometric systems.

In this experiment, students will apply the Discrete Fourier Transform (DFT) to an image, analyze its frequency spectrum, and implement Low-Pass and High-Pass Frequency Filters to observe their effects on image quality and feature extraction.

**Prerequisites**

- Basic knowledge of Digital Image Processing
- Understanding of spatial and frequency domains
- Familiarity with Python programming
- Python libraries: OpenCV (cv2), NumPy, and Matplotlib
- Jupyter Notebook, Google Colab, or Visual Studio Code

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
```

```
uploaded = files.upload()
```

[Choose Files](#) Screenshot... 004515.png

**Screenshot 2026-08-28 004515.png**(image/png) - 610254 bytes, last modified: 8/28/2026 - 100% done  
Saving Screenshot 2026-08-28 004515.png to Screenshot 2026-08-28 004515 (1).png

```
filename = list(uploaded.keys())[0]

img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

plt.imshow(img, cmap='gray')
plt.title("Original Grayscale Image")
plt.axis("off")
plt.show()
```

Original Grayscale Image

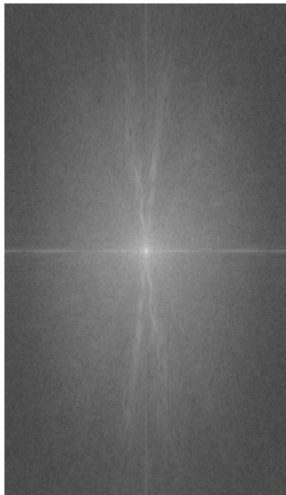


```
dft = np.fft.fft2(img)
dft_shift = np.fft.fftshift(dft)

magnitude_spectrum = 20 * np.log(np.abs(dft_shift) + 1)

plt.imshow(magnitude_spectrum, cmap='gray')
plt.title("Magnitude Spectrum")
plt.axis("off")
plt.show()
```

Magnitude Spectrum



```
rows, cols = img.shape
crow, ccol = rows // 2, cols // 2

radius = 50

mask_lpf = np.zeros((rows, cols), np.uint8)
cv2.circle(mask_lpf, (ccol, crow), radius, 1, -1)

filtered_lpf = dft_shift * mask_lpf

img_lpf = np.fft.ifft2(np.fft.ifftshift(filtered_lpf))
img_lpf = np.abs(img_lpf)

plt.imshow(img_lpf, cmap='gray')
plt.title("Low-Pass Filtered Image")
plt.axis("off")
plt.show()
```

Low-Pass Filtered Image



```
mask_hpf = np.ones((rows, cols), np.uint8)
cv2.circle(mask_hpf, (ccol, crow), radius, 0, -1)

filtered_hpf = dft_shift * mask_hpf

img_hpf = np.fft.ifft2(np.fft.ifftshift(filtered_hpf))
img_hpf = np.abs(img_hpf)

plt.imshow(img_hpf, cmap='gray')
plt.title("High-Pass Filtered Image")
plt.axis("off")
plt.show()
```

High-Pass Filtered Image



```
plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(magnitude_spectrum, cmap='gray')
plt.title("Magnitude Spectrum")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(img_lpf, cmap='gray')
plt.title("Low-Pass Filter")
plt.axis("off")

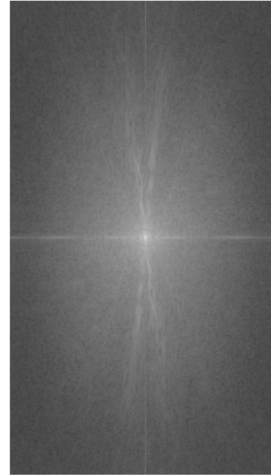
plt.subplot(2, 2, 4)
plt.imshow(img_hpf, cmap='gray')
plt.title("High-Pass Filter")
plt.axis("off")
```

```
plt.tight_layout()
plt.show()
```

Original Image



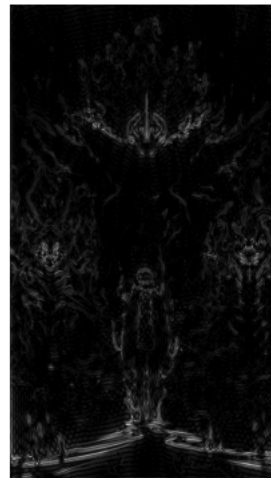
Magnitude Spectrum



Low-Pass Filter



High-Pass Filter



9. Analyze the impact of frequency domain filtering on image smoothness, edge enhancement, and noise removal.

#### ANS Low-Pass Filtering

Low-pass filtering suppresses high-frequency components.

##### Effects:

Reduces noise. Smooths the image. Removes fine details. May slightly blur sharp edges.

#### High-Pass Filtering

High-pass filtering suppresses low-frequency components.

##### Effects:

Enhances edges. Highlights fine details. Suppresses smooth background regions. Can also enhance high-frequency noise.

#### Filter Radius

The radius controls the amount of filtering.

Smaller radius: Stronger filtering. Larger radius: More frequency components are retained.

#### 10. Observations and Advantages of Frequency Domain Filtering

##### Observations:

- Low-pass filtering makes the image smoother by removing high-frequency components and reducing noise.
- High-pass filtering enhances edges and fine details by removing low-frequency components.
- Frequency filtering allows specific frequency components to be selectively modified.
- The filter size or radius affects the amount of smoothing and detail enhancement.

##### Advantages in Practical Computer Vision:

- Useful for noise removal and image smoothing.

- Helps in edge and feature enhancement.
- Useful for image restoration and enhancement.
- Can improve the quality of images before object detection and recognition.
- Used in applications such as medical imaging, satellite imaging, face recognition, and industrial inspection.

1. What is the Fourier Transform? Why is it important in digital image processing?

**ans:** The Fourier Transform converts an image from the spatial domain into the frequency domain. It represents the image using different frequency components. It is important for image filtering, noise reduction, smoothing, and edge enhancement.

2. Differentiate between the spatial domain and the frequency domain.

**ans:** Spatial Domain: Image processing is performed directly on the pixels of an image. Frequency Domain: Image processing is performed on the frequency components obtained using the Fourier Transform.

3. What is the significance of the Discrete Fourier Transform (DFT) in image processing?

**ans:** The DFT converts a digital image from the spatial domain to the frequency domain. It helps identify low and high-frequency components and is widely used for image filtering, enhancement, restoration, and analysis.

4. Explain the purpose of shifting the zero-frequency component to the center.

**ans:** The zero-frequency component is shifted to the center of the spectrum using `fftshift()`. This makes the frequency spectrum easier to visualize and analyze, with low frequencies at the center and high frequencies toward the edges.

5. Compare Low-Pass Frequency Filters and High-Pass Frequency Filters with suitable applications.

**A Low-Pass Filter (LPF)** allows low-frequency components to pass and removes high-frequency components. It is mainly used for image smoothing and noise reduction.

**A High-Pass Filter (HPF)** allows high-frequency components to pass and removes low-frequency components. It is mainly used for edge detection, sharpening, and detail enhancement.

6. What is the role of the Inverse Fourier Transform (IDFT) in image reconstruction?

**ans:** The Inverse Fourier Transform (IDFT) converts the processed frequency-domain data back into the spatial domain. It is used to reconstruct the final filtered image after applying frequency-domain operations.

7. Why is frequency domain filtering preferred for certain image enhancement tasks?

**ans:** Frequency domain filtering provides direct control over different frequency components of an image. It makes it easier to remove noise, smooth images, enhance edges, and modify specific frequency ranges.

8. Mention four real-world applications where Fourier Transform is used in computer vision and image analysis.

**Four real-world applications are:**

- Medical image processing – MRI and CT image analysis.
- Face recognition – Extracting useful image features.
- Satellite image processing – Improving and analyzing satellite images.
- Object detection – Enhancing important features and edges.

9. Compare frequency domain filtering with spatial domain filtering based on computational efficiency and practical applications.

**ans:** Spatial domain filtering works directly on image pixels and is generally simple and efficient for small filters. It is commonly used for basic operations such as blurring, sharpening, and edge detection.

Frequency domain filtering requires DFT and IDFT, so it can have more computational overhead for small operations. However, it can be more efficient for large filters and complex frequency-based operations and provides better control over frequency components.

10. How does frequency domain filtering improve the performance of image restoration and feature extraction techniques?

**ans:** Frequency domain filtering improves image restoration by removing unwanted noise and frequency components. It also enhances important features such as edges, boundaries, and fine details, making feature extraction and further image analysis more effective.